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Sex-specific risk assessment of PFHxS using a physiologically based pharmacokinetic model

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Abstract
Perfluorohexanesulfonate (PFHxS), which belongs to the group of perfluoroalkyl and polyfluoroalkyl substances (PFASs), has been extensively used in industry and subsequently detected in the environment. Its use may be problematic, as PFHxS is known to induce neuronal cell death, and has been associated with early onset menopause in women and with attention deficit/hyperactivity disorder. Due to these impending issues, the aim of this study is to develop and evaluate a physiologically based pharmacokinetic (PBPK) model for PFHxS in male and female rats, and apply this to a human health risk assessment. We conducted this study in vivo after the oral or intravenous administration of PFHxS in male (dose of 10 mg/kg) and female rats (dose of 0.5–10 mg/kg). The biological samples consisted of plasma, nine tissues, urine, and feces. We analyzed the sample using ultra-liquid chromatography coupled tandem mass spectrometry (UPLC–MS/MS). Our findings showed the tissue-plasma partition coefficients for PFHxS were highest in the liver. The predicted rat plasma and tissue concentrations using a simulation fitted well with the observed values. We extrapolated the PBPK model in male and female rats to a human PBPK model of PFHxS based on human physiological parameters. The reference doses of 0.711 μg/kg/day (male) and 0.159 μg/kg/day (female) and external doses of 0.007 μg/kg/day (male) and 0.006 μg/kg/day (female) for human risk assessment were estimated using Korean biomonitoring values. This study provides valuable insight into human health risk assessment regarding PFHxS exposure.

Keywords PBPK · Perfluorohexanesulfonate (PFHxS) · Human health risk assessment · Rat

Abbreviations

$AUC_{0-\infty}$	Area under the concentration–time curve from zero to infinity
CL	Clearance
C_{max}	Maximum plasma concentration
C_{ss}	Steady-state concentration
F_r	Free fraction in plasma
IV	Intravenous
K_p	Tissue-to-plasma partition coefficient
K_{st}	Rate constant to storage compartment
K_t	Transporter affinity constant
K_u	Urinary elimination rate constant
MOE	Margin of exposure
NOAEL	No-observed-adverse-effect level
PBPK	Physiologically based pharmacokinetic model
PFASs	Perfluoroalkyl and polyfluoroalkyl substances
PFHxS	Perfluorohexanesulfonate
PFOA	Perfluorooctanoic acid
PFOS	Perfluorooctane sulfonate
PK	Pharmacokinetic
POD	Points of departure
RfD	Reference dose
$t_{1/2}$	The elimination half-life
T_m	Transporter maximum
UF	Uncertainty factor
UF_A	Uncertainty factor for interspecies extrapolation from rats to humans
UF_H	Uncertainty factor for inter-individual variability in humans

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UF_S	Uncertainty factor for subchronic to chronic extrapolation
UPLC-MS/MS	Ultra-liquid chromatography coupled tandem mass spectrometry
V_d	Volume of distribution

Introduction

Perfluorohexanesulfonate (PFHxS, Fig. 1) belongs to the group of perfluoroalkyl and polyfluoroalkyl substances (PFASs), and has been extensively used as a protective agent in paper, waterproof textiles, and fire-fighting foams. Because of its indiscriminate use, PFHxS has been detected globally in environments as well as in humans (Kato et al. 2011; Sinclair et al. 2006). For example, associations with PFHxS concentration and bone mineral density in osteoporosis patients were found in the National Health and Nutrition Examination Survey (NHANES) in 2009–2010 (Khalil et al. 2016). Gleason et al. (2015) assessed the association with PFHxS and alanine transferase in the NHANES for 2007–2008 and 2009–2010 combined. In China, the associations between PFHxS levels in cord serum and birth weight, birth length, and ponderal index were investigated by measuring the cord serum concentration of PFHxS (Shi et al. 2017). Another study in Korea found that PFHxS concentrations ranged from 0.26 to 2.46 ng/mL in the serum of children aged 5–13 years from Dae gu (Kim et al. 2014), and 1.35–1.75 and 2.19–2.29 ng/mL in the serum of women and men, respectively (MFDS 2009). The detection of PFHxS has also been compared by geographical area: Seoul (median = 0.54 ng/mL); Busan (median = 0.90 ng/mL); and Yeosu (median = 0.19 ng/mL) (Cho et al. 2015). Numerous publications reported on the gender difference of PFHxS in human serum. The human serum concentration of PFHxS in males was higher than that of females in Australians from 2002 to 2013 (Eriksson et al. 2017), Singaporeans (Liu et al. 2017), southern Chinese (Li et al. 2017), and human plasma collected from American Red Cross male and female blood donors (ages 20–69) at 6 regional blood collection centers (Olsen et al. 2017).

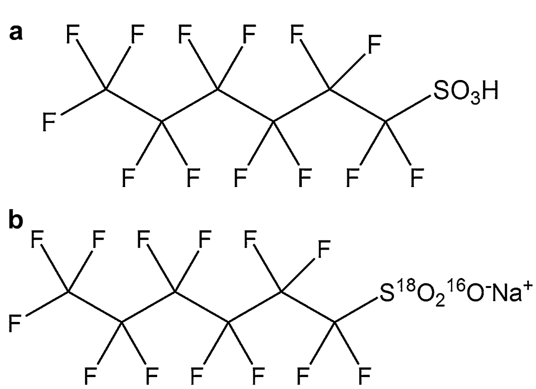

Fig. 1 Chemical structures of **a** PFHxS and **b** MPFHxS as internal standard

The detection in the Korean population is concerning as the toxicity studies have reported that PFHxS induces apoptosis of dopaminergic neuronal cells and cerebellar granule cells (Lee et al. 2014a, b). High exposure to PFHxS has also been strongly associated with attention deficit/hyperactivity disorder (ADHD) in children (Stein and Savitz 2011) and with low total cholesterol levels (Tillett 2010). The dectection of PFHxS is also correlated with gender-specific issues. For example, women with high blood levels of PFHxS have higher rates of hysterectomies than women with lower levels (Taylor et al. 2014). High levels may also lead to early onset of menopause, but the underlying mechanism for this is not known (Taylor et al. 2014). Despite these toxicity reports, the use of PFHxS has not been regulated.

Several pharmacokinetic (PK) studies have been carried out for PFHxS. The elimination half-lives ($t_{1/2}$) of PFHxS in monkeys, mice, male rats, and female rats were approximately 4 months, 1 month, 1 month, and 2 days, respectively (Sundstrom et al. 2012). Recently, PK studies showed significant gender differences between the PK parameters of PFHxS, in which $t_{1/2}$ was 20.7–26.9 days in male rats and 0.9–1.7 days in female rats (Kim et al. 2016). In humans, $t_{1/2}$ was approximately 8.5 years and showed no significant gender difference between male (8.2 years) and female (12.8 years) workers who retired from fluorochemical production industries (Olsen et al. 2007). On the contrary, Zhang et al. (2013) showed a difference in the $t_{1/2}$ between male (35 years) and female (7.7 years) humans. They estimated the elimination $t_{1/2}$ of PFHxS from paired blood and urine samples (Zhang et al. 2013). Han et al. (2012) described that PFASs showed the species differences for renal tubular secretion and reabsorption, which are mediated via organic anion transporters (OATs), and summarized the renal OATs of mouse, rat, and human. Oat5 is rat specific and OAT4 is human specific. In the membrane of the proximal tubular cells, Oat2 in rodents is located in the apical site, whereas human OAT2 is expressed in the basolateral site (Han et al. 2012). In addition, the Oatp1a1 in rats (Yang et al. 2009) and OAT4 and URAT1 in human (Yang et al. 2010) have been suggested to play a role of renal tubular reabsorption for PFOA.

During the past decade, the physiologically based pharmacokinetic (PBPK) model for perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS) in monkeys was developed and extrapolated to humans for use in exposure assessments (Loccisano et al. 2011). PBPK models of PFOA and PFOS in adult male and female rats were also developed and compared to characterize the PKs of PFOA and PFOS (Loccisano et al. 2012a). The PBPK model of

PFOA and PFOS in pregnant, lactating, fetal, and neonatal rats was developed to evaluate the placental and lactational PKs of PFOA and PFOS (Loccisano et al. 2012b). In addition, a PBPK model was used to explain the epidemiologic association between the serum level of PFASs (PFOA and PFOS) and the early onset of menopause (Ruark et al. 2017). A PBPK model has also been used to explain the relationship between serum PFASs levels and delayed menarche (Wu et al. 2015). Only one publication has reported on a PBPK model of PFHxS to estimate the prenatal and postnatal exposure of PFHxS, PFOA, and PFOS using PK model simulation (Verner et al. 2016). To the best of the authors' knowledge, this is the first report of a PBPK model considering the sex-specificity of PFHxS. In the PBPK model, the free fraction of PFASs is an important parameter for describing the kinetics of chemicals. Only the free form of chemicals in plasma is transferred to tissues and reabsorbed back to the kidney from the filtrate compartment (Andersen et al. 2006; Tan et al. 2008). However, all the abovementioned PBPK models used the free fraction of PFOA and PFOS through the model fitting, and in the models, PFOA and PFOS are considered to be 97% bound to plasma proteins (Han et al. 2003; Jones et al. 2003). Therefore, it is necessary to directly measure the free fraction of PFHxS in plasma to adequately describe the PBPK model.

A report was presented on a human health risk assessment for levels of PFOA and PFOS in the atmosphere of Shenzhen, China (Liu et al. 2015) as well as in three distinct European populations (Ludwicki et al. 2015). While these reports only discussed the presence of PFOA and PFOS, Borg et al. (2013) published a health risk assessment of 17 PFCs including PFHxS in a Swedish population. In the three reports mentioned above, the hazard quotient (HQ) was used as a tool for a health risk assessment (Borg et al. 2013). In 2015, the Agency for Toxic Substances and Disease Registry (ATSDR) released the minimal risk levels of PFOA (0.02 µg/kg/day) and PFOS (0.03 µg/kg/day) for hepatic end point (ATSDR 2015).

Moreover, the health risk assessment of PFHxS using a PBPK model has not been reported until now. The World Health Organization (WHO) presented a guide for applying the PBPK model for risk assessment (WHO 2010). In 2015, the Organisation for Economic Co-operation and Development (OECD) presented risk reduction approaches for PFASs across various countries to reduce their influence on the environment and on human health and to support a global transition toward the use of more benign alternatives (OECD 2015).

The goals of this study were to develop and evaluate a PBPK model for PFHxS in male and female rats and to apply the model to the sex-specific risk assessment for human exposure. The developed PBPK model was extrapolated to a male and female human model, and used to estimate the risk assessment of PFHxS through simulation. This study provides valuable insight into human health risk assessment of PFHxS exposure and could be used to compare the risk assessments for male and female humans.

Materials and methods

Materials and reagents

The reference standard of K^+PFHxS (purity > 99.5%) was supplied from Sigma–Aldrich (St Louis, MO, USA). Sodium perfluoro-1-hexane[$^{18}O_2$] sulfonate (MPFHxS, purity > 98%), used as an internal standard (IS), was obtained by Wellington Laboratories Inc. (Guelph, ON, Canada). All chemicals used were of HPLC grade or better quality. Distilled water was produced by an Elga Purelab Option-Q system (Elga LabWater, Marlow, UK) for this study.

Animals

Female and male Sprague–Dawley rats (8–12 weeks, 240–260 g) were obtained from Damul Sciences (Daejeon, Republic of Korea). All animals were separately housed in a metabolic cage in a ventilated room that was maintained at a controlled temperature of 23 ± 2 °C, a relative humidity of 50 ± 10%, and a 12 h light/dark cycle. Rats were used for experiments after one week of acclimation. All our experiments were carried out in the CHA laboratory animal research center and were approved by the Institutional Animal Care and Use Committee (IACUC, no. IACUC140047).

Pharmacokinetic study in rats

We divided the female rats into six groups according to administration doses or routes. Considering the short $t_{1/2}$ of PFHxS in female rats (Kim et al. 2016; Sundstrom et al. 2012), the female rats were administered intravenously (IV) to estimate the dose-dependent linearity and the partition coefficient (K_p) values. In four of the groups, PFHxS was administered at 0.5, 1, 4, and 10 mg/kg doses into a tail vein ($n = 5$/group). For the development and validation of the PBPK model, two groups of female rats were orally administered with doses of 1 and 4 mg/kg ($n = 5$/group) considered exposure routes. For the male group, the PK study was designed to obtain the minimum experimental data for developing the PBPK model. The male rats were divided into two groups, those given the PFHxS via oral administration and those given the PFHxS by an IV injection at 10 mg/kg ($n = 5$/group).

Blood samples were obtained via the jugular vein at 0.5, 1, 2, 4, 6, 12, 24 (h), and 2, 3, 5, 7, 10, 14 days for female rats, and at 0.33, 0.67, 1, 2, 6, 12, 24 (h), and 1, 2, 3, 5, 7, 10, 14 days for male rats. Immediately after taking the blood samples, they were centrifuged (7000 rpm, 10 min). Plasma was then taken from the blood samples and stored at − 80 °C until assay. Heparin was used as an anticoagulant. After the last blood sample was taken, all rats were killed, and the tissue samples of the brain, heart, liver, lung, spleen, kidney, gastrointestinal (GI) tract, adipose tissue, and muscle were collected. The collected tissues were accurately weighed and kept in a freezer at − 80 °C until the analysis. We used the metabolic cages to obtain urine and feces. After the total mass of feces and total volume of urine were measured, they were stored at − 80 °C until assay. We calculated the PK parameters using non-compartmental and compartmental analyses with WinNonlin® software (version 6.4, Pharsight®, Certara™ Company, Princeton, NJ, USA).

Determination of PFHxS in rat plasma and tissues

The quantitation of PFHxS in the biological sample was measured in an Acquity UPLC® system coupled with a mass spectrometer (Xevo TQ-S, Waters Corp., Milford, MA, USA). Chromatography was performed using a Halo C_{18} column (2.1 mm × 50 mm, 2.7 μm particle size) at a temperature of 40 ± 0.5 °C. The mobile phase was composed of 2 mM ammonium acetate in water and acetonitrile, with a gradient condition as follows: 0–0.5 min, 15% of B; 0.5–1.2 min, 15–90% of B; 1.2–2.4 min, 90% of B; 2.4–2.5 min, 90–15% of B; 2.5–4.0 min, 15% of B. The flow rate was at 0.2 mL/min. The mass spectrometer was operated with the electrospray ionization interface in the negative ion mode. Multiple reaction monitoring (MRM) transitions were m/z 398.9 → 79.6 for PFHxS and m/z 402.9 → 102.9 for MPFHxS for IS, respectively. The optimized parameters for the capillary voltage, cone voltage, and collision energy were 2.8 kV, 20 V, and 35 eV, respectively. The Masslynx 4.1 software (Waters Corp., Milford, MA, USA) was used for data acquisition and analysis.

The standard calibration range of PFHxS was 5–5000 ng/mL in rat plasma and tissues. The samples were extracted using a combination of protein precipitation and liquid–liquid extraction. The plasma samples were diluted by distilled water with the five times of sample volume and the tissues were homogenized (Ultra-Turrax®, IKA® T10 basic) in distilled water with five times of sample amount. Then, 200 μL of each sample was added to 10 μL of the IS solution (100 ng/mL in 50% methanol) and 600 μL of acetonitrile. The solution was then vortex-mixed for 1 min and centrifuged at 15,000×g for 5 min. The resulting supernatant of 600 μL was transferred to a clean tube to which was added 600 μL of methylene chloride. After vortex-mixing for 3 min and centrifuging at 15,000×g for 5 min, 600 μL of the organic layer was transferred into a new tube and was dried under a gentle stream of nitrogen gas at 50 °C. The residue was reconstituted with 30% acetonitrile, and 2 μL of supernatant was injected into UPLC-MS/MS. The samples exceeding the calibration range were reanalyzed after dilution. The method developed for PFHxS detection was validated in accordance with the Guidance for Industry: Bioanalytical Method Validation (FDA 2013).

Protein binding assay in rat and human plasma

Protein binding assay was examined in rat and human plasma for both male and female. We used the Micropartition System Starter kit with YMT membrane (Amicon, Denver, CO, USA). The quintuplicate plasma containing the PFHxS with 10 μg/mL concentration was placed in the sample reservoir and centrifuged at 1000 × g for 10 min (n = 5). A protein-free ultrafiltrate was collected, of which the amount of the initial volume of plasma ranged between 20 and 30%. The concentration of PFHxS was measured by the abovementioned UPLC-MS/MS. The percentage of plasma protein binding (1) was calculated from the concentration in the ultrafiltrate (C_{UF}) and plasma in the sample reservoir (C_{total}) using Eq. (1) as follows:

$$\text{Protein binding (\%)} = \left(C_{total} - C_{UF}\right)/C_{total} \times 100 \quad (1)$$

Development of PBPK model

We considered the PK and pharmacodynamic properties of the general PFASs to develop a PBPK model for PFHxS. The model comprised compartments of blood, liver, filtrate, storage, and tissue. These compartments consisted of organs and tissues that were interconnected in the body via blood flow.

We adopted the use of tissue compartments for our model because of their relevance in toxicokinetics in the body: the brain as the target organ of the neurotoxic effects, the GI tract as an absorption site, and adipose tissue as a site of the accumulation in lipophilic tissues after exposure to PFHxS. The liver was included, considering the possibility of enterohepatic circulation (Harada et al. 2007), as well as an accumulation site in which the PFASs interacted with the rat liver fatty acid-binding protein (Luebker et al. 2002; Sheng et al. 2016). PFHxS was shown to preferentially distribute to the liver of the rodents and activated the peroxisome proliferator-activated receptor α (PPARα), as well as other nuclear receptors, including constitutive activated receptor (CAR), PPARγ, and estrogen receptor α (ERα) (Rosen et al. 2017). The muscle was included as an independent compartment in order to retain the PFASs in other tissues. The filtrate compartment was included in the model to explain the role of renal reabsorption in a saturable transport process that can reabsorb back into the kidney compartment. The free fraction of chemicals in plasma is assumed to be transferred into the tissues. The menstrual blood loss was only adopted for female human as one of the excretion routes, and the PFHxS was excreted via the filtrate compartment to storage and then to urine. Figure 2 shows the structures of the PBPK model for PFHxS in rats. The physiological parameters (such as tissue volume and blood flow) and the chemical properties of PFHxS were used to develop the rat PBPK model to extrapolate a human model for risk assessment. Differential mass balance equations were derived for individual tissues, and the equations listed in the Appendix show the mass balance of PFHxS in the blood, the oral compartment, the liver and kidney compartments, the filtrate compartment, the storage compartment, and tissues after oral and IV administration.

Model parameterization

The values of the physiological parameters used to develop the PBPK model were sourced from the related literature (Davies and Morris 1993; Igari et al. 1983) and are listed in Table 1. The K_p values for PFHxS in male and female rats were obtained from the experimental data based on an in vivo study. The K_p values of each organ were calculated from the ratio of concentration of plasma and each organ at 14 days after IV injection at a dose of 0.5–10 mg/kg

Table 1 Physiological parameters used in the PBPK model for PFHxS

Tissue[a]	Rat		Human	
	Tissue volume (mL)	Blood flow (mL/h)	Tissue volume (mL)	Blood flow (mL/h)
Blood	20.4	2580	5400	379,800
Heart	0.8	234	300	14,400
Brain	0.7	18	1500	42,000
Lung	1.0	2580	1200	379,800
Adipose tissue	10.0	24	12,200	15,600
Muscle	122.0	450	30,000	45,000
Kidney	1.8	520	300	74,400
Filtrate[b]	0.18	260	30	14,880
G.I tract	10.0	450	2400	66,000
Liver	8.0	828	1500	87,000
Spleen	1.3	36	192	4620

[a]Tissue volume and blood flow in rats and humans were obtained from Davies and Morris (1993) and Igari et al. (1983)

[b]Tissue volume and blood flow of filtrate are referred to Loccisano et al. (2012a) for rats and Loccisano et al. (2011) for humans. Filtrate volume was assumed a 10% of kidney volume in both species, and blood flow was 50% (rats) and 20% (humans) of kidney blood flow

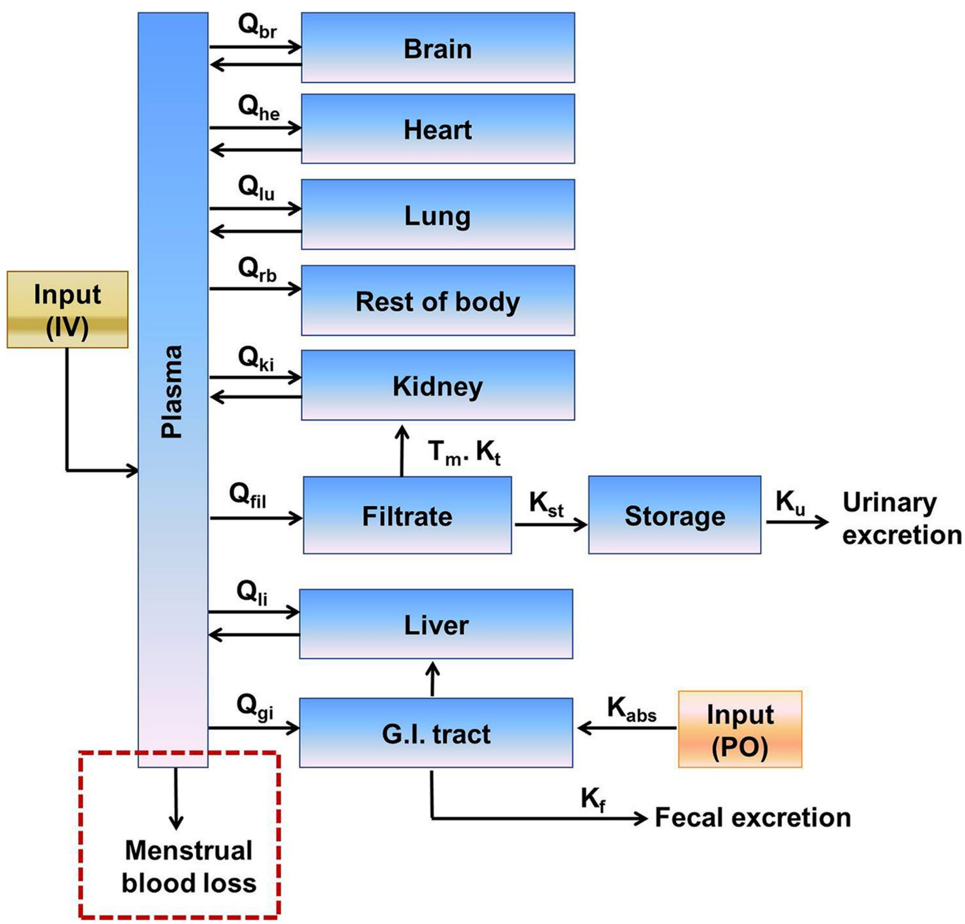

Fig. 2 Structure of the PBPK model for PFHxS in rats and humans. PFHxS can be reabsorbed into the kidney with transporter maximum (T_m) and transporter affinity constant (K_t). K_s indicates a rate constant; K_{st}, the rate constant to the storage compartment; K_u, the urinary elimination rate constant; K_f, the transfer rate constant from the GI tract to fecal elimination; K_{abs}, the oral absorption rate constant. Q_s refers to the blood flows between plasma and tissues, except for Q_{fil}, which is a clearance from the plasma to the filtrate compartment. Menstrual blood loss (dot square) is only applicable to female humans.

administered to rats. The equation used to estimate the K_p (2) of each organ was as follows:

$$K_p = \frac{C_{tissue}}{C_{plasma}} \tag{2}$$

When the chemical is administered orally, the dose enters the liver compartment, except for a small unabsorbed fraction that is eventually found in the feces. In this model, the saturable renal reabsorption was performed in the filtrate compartment to describe the kinetics of the PFHxS. A PBPK model of PFOA and PFOS with a structure similar to that of PFHxS in monkeys was referred to describe the renal tubular reabsorption (Loccisano et al. 2011). The transporter maximum (T_m), the transporter affinity constant (K_t), the rate constant to storage compartment (K_{st}), and the free fraction of PFHxS in plasma (F_r) were used to describe the model. The T_m, K_t, and K_{st} for PFHxS were estimated by fitting the plasma and urine concentration data in rats. The F_r of PFHxS was obtained from the plasma protein binding assay. The urinary elimination rate constant (K_u) of PFHxS was estimated by fitting the urine concentration from the experimental data.

The PBPK model was fitted and simulated using the Berkeley Madonna V 8.3.14 software (Berkeley, CA, USA), which is one of the types of simulation software suggested by the United States Environmental Protection Agency (US EPA) as an approach for applying the PBPK model in risk assessment (EPA 2006).

Model validation

Model validation was conducted to compare the experimental data and simulated concentrations from oral dose routes. The adequacy of the PBPK model in male and female rats was estimated based on whether the model could describe the PK data of the PFHxS in male and female rats. The PBPK model was evaluated according to the values observed for the PFHxS concentrations in plasma, tissues, and urine, compared to the predicted concentration at different time points. This evaluation was visually examined and found to be consistent across the simulation and concentration obtained from the biological samples. Overall, the experimental data and the model simulation should be in good agreement. To validate the PBPK model, the male and female rats were both administered 1 and 4 mg/kg doses ($n = 5$/group) via oral administration considering the exposure route. Plasma and tissue samples were collected according to our experimental design. The plasma and tissue samples were kept at − 80 °C until analysis.

A normalized sensitivity analysis was performed on the developed PBPK model of PFHxS for risk assessment. The influence of each biochemical parameter was examined by calculating the area under the concentration–time curve (AUC) after increasing each parameter by 1%. Each parameter was increased individually. The normalized sensitivity coefficients (3) were calculated using the following equation:

$$\text{sensitive coefficient} = \frac{A-B}{B} \Big/ \frac{C-D}{D} \tag{3}$$

where A is the AUC resulting from the 1% increase in parameter value, B is the AUC resulting from the original parameter value, C is the parameter value increased by 1%, and D is the original parameter value. Plasma AUC predictions were performed at 10 mg/kg dose for male rats and at 1 and 4 mg/kg oral doses for female rats after oral administration.

Extrapolation to PBPK model in humans

The PBPK model structure for PFHxS in humans was consistent with that in rats. The rat PBPK model was extrapolated to humans, considering the interspecies differences of physiological and chemical-specific parameters. The physiological parameters (tissue volume, blood flow, and K_p) in rats were replaced with the corresponding physiological parameters for humans (Table 1). The average body weight (BW) of rats and humans was assumed to be 0.25 and 70 kg, respectively, and all parameters of the same tissue for rats and humans were assumed to be identical. We estimated k_{human} using the following allometric equation (Hu and Hayton 2001; Kagan et al. 2011; Sharma and McNeill 2009).

$$k_{human} = k_{rat} \cdot \left(\frac{BW_{human}}{BW_{rat}} \right)^{-0.25} \tag{4}$$

where k_{human} and k_{rat} are the rate constants in humans and rats, respectively. BW_{human} and BW_{rat} refer to the body weight in humans and rats, respectively; − 0.25 is the allometric exponent. The simulation of the PBPK model in male and female rats was performed at a no-observed-adverse-effect level (NOAEL) dose of 1 mg/kg, as a repeated administration (Butenhoff et al. 2009).

Human health risk assessment

Viberg et al. (2013) reported that the neonatal exposure at doses of 0.61, 6.1, and 9.2 mg/kg PFHxS in mice could induce a developmental neurotoxicant, thereby affecting the cholinergic system and cognitive function (Viberg et al. 2013). Because our study suggested the PBPK model in rats, the points of departure (PODs) were based on NOAELs via oral administration in rats, considering the exposure route of PFHxS. The NOAELs of PFHxS in rats were 10 mg/kg/day for immune/lymphoreticular toxicity, neurological toxicity, reproductive toxicity, developmental toxicity, and increased urea nitrogen in serum, and were 1–10 mg/kg/day for the organ hypertrophy (Butenhoff et al. 2009). As such, we selected a POD of 1 mg/kg/day for organ hypertrophy in male and female rats, considering their high sensitivity to the effect of PFHxS exposure.

The steady-state concentration (C_{ss}) of PFHxS in male and female rats was estimated by inputting the POD to the PBPK model in rats. The C_{ss} in rats was used to predict the C_{ss} in humans using an uncertainty factor (UF) with interspecies extrapolation. The uncertainty factor was $10^{0.5}$ for interspecies extrapolation from rats to humans (UF_A) (WHO 2010). The PODs in the male and female humans were obtained through the dose-metric approach, as suggested by the WHO (2010). The external dose was derived with an average plasma concentration from the repeated exposure of PFHxS using the human PBPK model. According to the guidance for risk assessment using the PBPK model by the US EPA (2006) and WHO (2010), we estimated the reference dose (RfD) (5) as follows:

$$\mathrm{RfD} = \mathrm{POD/UF} \tag{5}$$

where POD is a NOAEL in humans, and UF is the uncertainty factor related to extrapolations associated with the POD. The UF was set at 10 for inter-individual variability in humans (UF_H) (WHO 2010) and at 10 for subchronic to chronic extrapolation (UF_S) (Dourson and Stara 1983). The PFHxS concentration used to estimate the external dose was sourced from the human biomonitoring values of hazardous substances containing PFASs, as obtained by the Ministry of Food and Drug Safety (MFDS) in 2008 (MFDS 2009).

The margin of exposure (MOE) (6) was calculated as the ratio of the RfD to external exposure in humans, using the following equation:

$$\text{Margin of Exposure} = \frac{\text{Reference dose}}{\text{External exposure}} \tag{6}$$

The MOE, which is considered with various sources of uncertainty and variability, indicates the margin between the reference dose and the actual exposure.

Statistical analysis

We analyzed the data for statistical significance using a Student's t test with $p < 0.05$ indicating a significant difference. The Statistical Analysis System (SAS) software version 9 (SAS Institute) was used for the statistical analysis.

Results

Pharmacokinetic parameters

Table 2 summarizes the results of the PK parameters. The parameters were estimated using a non-compartment model after the IV or oral administration of PFHxS at doses of (0.5), 1, 4, and (10) mg/kg for female rats and at doses of 10 mg/kg for male rats (numbers in parenthesis indicate IV administration). Figure 3 shows the mean plasma concentration–time profiles of the PFHxS after IV or oral administration for male and female rats. For IV administration of 0.5–10 mg/kg PFHxS to female rats, the $AUC_{0-\infty}$ increased in a dose-dependent manner from 159.50 to 4575.70 μg·h/mL, whereas the $t_{1/2}$ of 38.71–48.92 h and CL of 2.19–3.09 mL/h/kg were constant regardless of the dose (Fig. 4). The $t_{1/2}$ of 819.25 h in male rats was at least 18 times longer than that of female rats (45.57 h), and the CL in females (2.55 mL/h/kg) was about 10 times higher than that of male rats (0.27 mL/h/kg). The $t_{1/2}$ and CL showed a significant gender difference in male and female rats. The bioavailability calculated by $AUC_{0-\infty}$ after IV and oral administration was estimated to be about 0.88–0.96; therefore, the gender difference of $t_{1/2}$ and CL is due to the distribution or the excretion instead of the absorption of PFHxS.

Figure 5 shows the tissue distribution of PFHxS in male rats at a dose of 10 mg/kg and female rats at doses of 0.5–10 mg/kg via IV administration. The K_p values of PFHxS were determined in nine tissues from the brain, heart, lung, kidney, liver, spleen, GI tract, adipose tissue, and muscle. In the female rats administered at doses of 0.5–10 mg/kg, the K_p value of each tissue was constant regardless of dosage. PFHxS was highly detected in the liver (0.127, male rats; 0.070, female rats), followed by the kidney (0.065, male rats; 0.047, female rats) and the lung (0.068, male rats; 0.044, female rats). The liver distribution of PFHxS in male rats was approximately 2 times higher than that in female rats. In other tissues, PFHxS was also more highly distributed in male rats than in female rats. The tissue distribution of PFHxS in the liver, lung, brain, spleen, GI tract, and adipose tissue showed significant gender differences between male and female rats ($p < 0.05$). However, the mean K_p value of nine tissues in female rats (0.027) was less than 2 times in male rats (0.045). These results show that the K_p value has a relatively low effect on the gender differences of $t_{1/2}$ and CL of PFHxS.

The urinary and fecal excretions of PFHxS were estimated after IV or oral administration to male rats (with a dose of 10 mg/kg PFHxS) and female rats (with a dose of 4 mg/kg PFHxS). Figure 6 shows the mean cumulative

Table 2 Pharmacokinetic parameters of PFHxS following oral or IV administration in male and female rats (mean ± SD)

Parameters	Female								Male	
	IV					Oral			IV	Oral
	0.5 mg/kg	1 mg/kg	4 mg/kg	10 mg/kg	Mean ± SD	1 mg/kg	4 mg/kg	Mean ± SD	10 mg/kg	10 mg/kg
$AUC_{0-\infty}$ (μg·h/mL)	159.50 ± 6.86	323.11 ± 56.67	1712.42 ± 88.60	4575.70 ± 155.59	NA	296.12 ± 19.35	1471.74 ± 57.37	NA	37542.29 ± 445.83	36124.72 ± 2218.63
C_{max} (μg/mL)[a]	3.31 ± 0.23	6.26 ± 0.30	24.24 ± 0.47	60.15 ± 2.50	NA	5.50 ± 0.09	17.45 ± 0.48	NA	32.99 ± 2.92	30.60 ± 3.94
$t_{1/2}$ (h)	41.08 ± 2.17	42.78 ± 1.09	38.71 ± 1.54	48.92 ± 2.10	45.57 ± 7.75	38.45 ± 2.01	40.45 ± 1.40	38.57 ± 7.68	819.25 ± 21.13*	817.94 ± 116.31
V_d (mL/kg)[b]	176.44 ± 11.20	191.05 ± 7.50	130.44 ± 5.54	153.90 ± 20.21	163.70 ± 20.28	187.34 ± 3.49	158.60 ± 7.78	170.65 ± 30.66	314.83 ± 23.21*	326.66 ± 19.81
CL (mL/h/kg)[b]	2.98 ± 0.35	3.09 ± 0.28	2.34 ± 0.26	2.19 ± 0.36	2.55 ± 0.49	3.38 ± 0.35	2.72 ± 0.34	3.09 ± 0.39	0.266 ± 0.024*	0.277 ± 0.048
F	NA	NA	NA	NA	NA	0.92 ± 0.12	0.88 ± 0.04	NA	NA	0.96 ± 0.08

NA not available

**P* < 0.05 between male and female

[a]The concentration is C_0 (μg/mL) after IV administration

[b]The value refers to the clearance (CL/*F*) and volume of distribution (V_d/*F*) after oral administration

excreted percentage of PFHxS in urine and feces. After IV administration, the percentage of PFHxS in urine and feces was 15.9 ± 4.2 and 1.3 ± 0.5%, respectively, in male rats and 39.1 ± 6.2 and 3.1 ± 0.08%, respectively, in female rats. After oral administration, they were 18.5 ± 3.6 and 2.8 ± 1.3%, respectively, in male rats and 36.8 ± 6.6 and 3.3 ± 0.67%, respectively, in female rats. These data show that female rats excreted more PFHxS mainly through the urine than the male rats.

Protein binding assay in rat and human plasma

The percentage of plasma protein binding was examined in rats and humans for both genders. The PFHxS was highly bound to plasma protein with 99.92 ± 0.004% in male rats and 99.93 ± 0.002% in female rats. In humans, the protein binding ratio was calculated at 99.98% in both genders (99.975% for male, 99.982% for female). No significant gender differences were found between male and female ($p < 0.05$).

Development and validation of PBPK model in rats

The biochemical-specific parameters were fitted based on the PBPK model in rats after IV or oral administration at doses ranging from 0.5 to 10 mg/kg. Table 3 summarizes each parameter. The parameters of T_m, K_t, and K_u were 85,300 μg/h/kg, 15,851 μg/L, and 0.250 h^{-1} in male rats, respectively, and 910 μg/h/kg, 15,851 μg/L, and 0.591 h^{-1} in female rats, respectively. Model fitting was carried out based on the changes of the key parameters that were unknown, and the simulation line and experimental values were visually compared. The PBPK model was constructed to simulate the experimental data obtained from the plasma and tissues in male (a dose of 10 mg/kg) and female (a dose of 0.5–10 mg/kg) rats exposed to a single administration of PFHxS with the physiological parameters and K_p values. Figures 7 and 8 show the predicted and observed individual concentrations of PFHxS in plasma and tissues after oral administration to male and female rats. The concentrations of PFHxS from the model simulations in male and female rats corresponded to the measured concentrations in the plasma, liver, kidney, heart, lung, and urine. The model included the GI tract as an absorption compartment and filtrate compartment for describing the renal reabsorption. The brain, known to be sensitive to neurotoxicity, was included in the PBPK model, even though the concentration of PFHxS in the brain was very low. The other tissues, such as muscle, spleen, and adipose tissue, were classified as belonging to the rest of body because of the low level of PFHxS in tissues. The predicted concentrations in each tissue fitted the experimental data well.

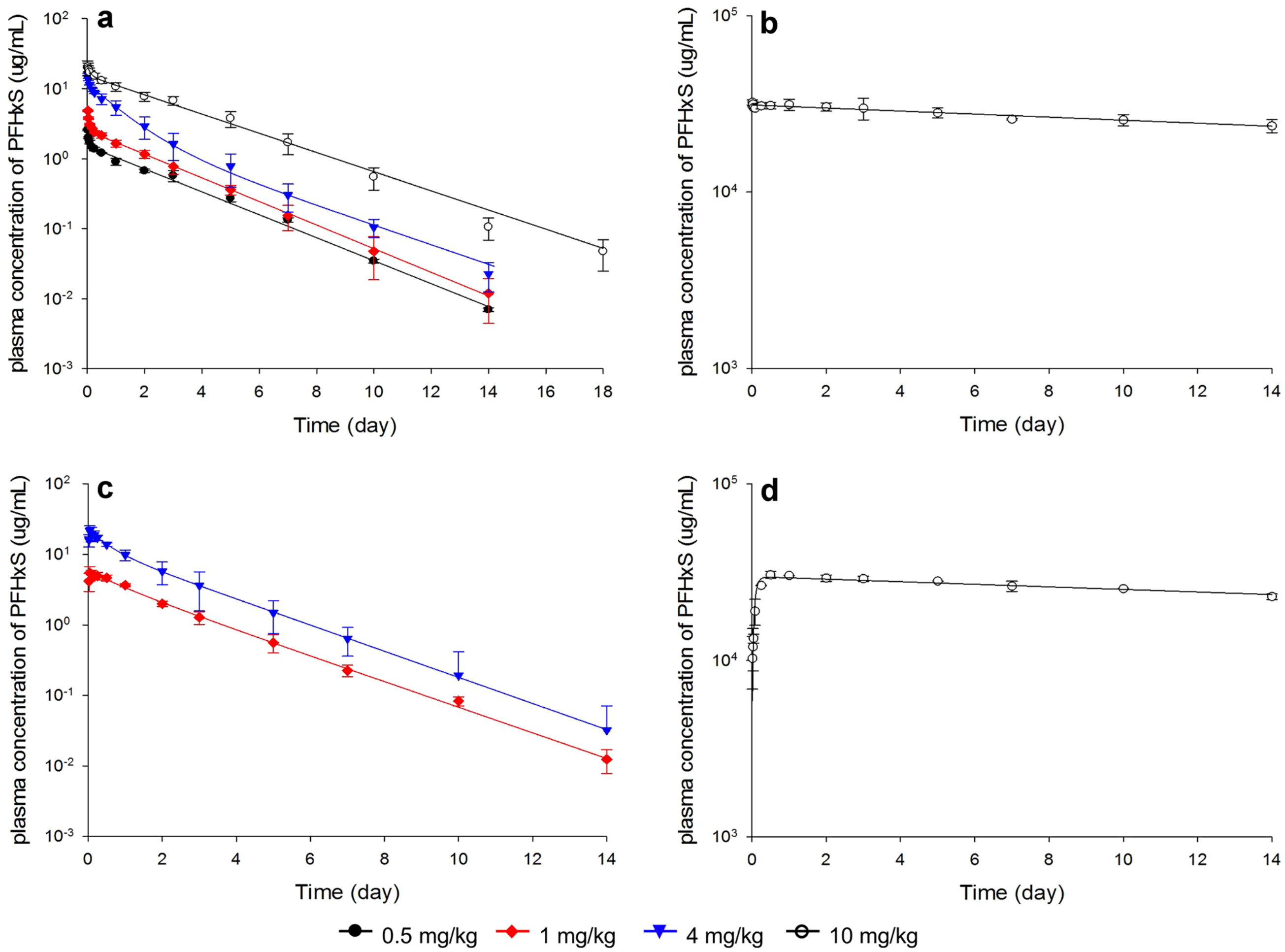

Fig. 3 Mean plasma concentration–time profile for PFHxS after IV and oral administration to male (right) and female (left) rats. The female rats were **a** injected via IV at doses of 0.5, 1, 4, and 10 mg/kg PFHxS and **c** orally administered at doses of 1 and 4 mg/kg PFHxS. Male rats were given a dose of 10 mg/kg PFHxS via **b** IV and **d** oral administration. Each value represents the mean ± SD (*n* = *5*)

The normalized sensitivity coefficients for the model parameters of PFHxS in rats are shown in Fig. 9. Any dose-dependent differences in female rats were not shown in the model sensitivity, but significant gender dependencies were observed. PFHxS was primarily dependent on the T_m (0.75), K_t (− 0.71), Q_{fil} (− 1.02), F_r (− 1.02), and K_{st} (− 0.71) for male rats, and Q_{fil} (− 0.59), F_r (− 1.05), and K_f (− 0.70) for female rats. The other parameters were less than 0.01 in male (K_u, K_f, and K_{abs}) and female (T_m, K_t, K_u, and K_{abs}) rats. PFHxS was influenced by parameters of Q_{fil} and F_r regardless of gender.

Extrapolation to the PBPK model in humans

The PBPK model structure of PFHxS in humans was the same as that for rats (Fig. 2). Figures 10 and 11 show the predicted concentration–time profiles of plasma, tissues, and urine for PFHxS in male and female humans. We ranked the PFHxS exposure in tissues as follows: liver > kidney and lung > heart > rest of body. The PK parameters of PFHxS in humans were estimated using a predicted concentration–time profile. The $AUC_{0-\infty}$ was 76,528 µg·h/mL after the oral administration of 10 mg/kg to male humans. The $AUC_{0-\infty}$ of female humans was calculated from 3516 to 70,320 µg·h/mL after the oral administration of 0.5–10 mg/kg and showed dose dependency.

Human health risk assessment

We used the rat NOAEL for PFHxS 1 mg/kg/day (as the most sensitive POD) to conduct the human health risk assessment based on the assumption of repeated daily exposure. The POD in humans was estimated to be 71.09 µg/kg/day for males and 15.89 µg/kg/day for females using a dose-metric approach for the external dose with an average plasma concentration derived from the human PBPK model and using UF_A of $10^{0.5}$. The reference dose was 0.711 µg/kg/day in males and 0.159 µg/kg/day in

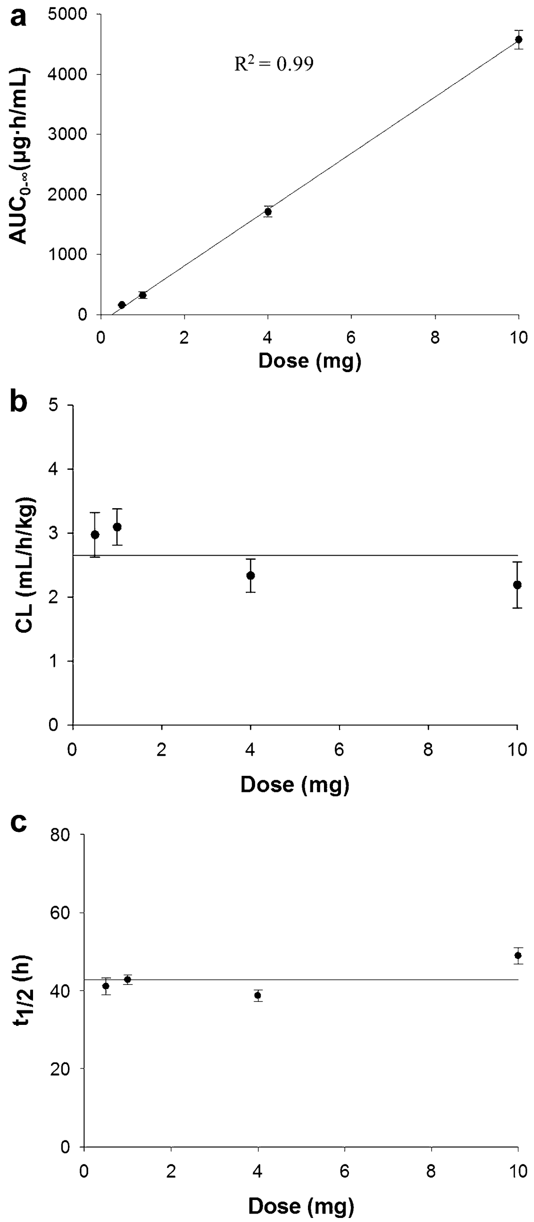

Fig. 4 Dose relationship with **a** $AUC_{0-\infty}$, **b** CL, and **c** $t_{1/2}$ of PFHxS in female rats after IV administration at a dose of 0.5–10 mg/kg. Each value represents the mean ± SD ($n = 5$)

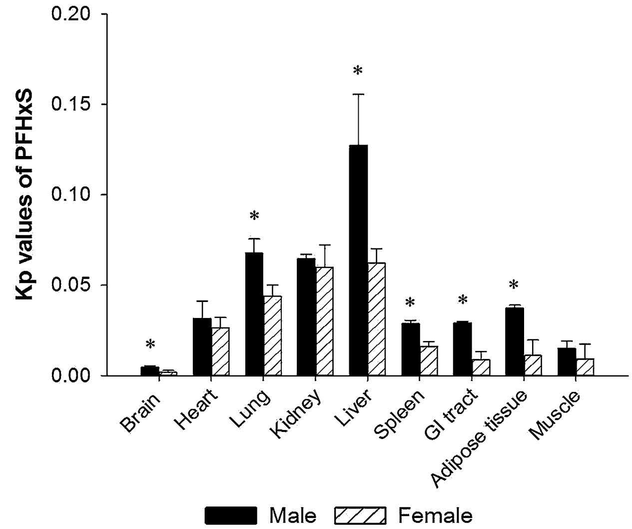

Fig. 5 Partition coefficient (K_p) of PFHxS in male and female rats. The K_p values were obtained from an in vivo study in which male rats were administered a dose of 10 mg/kg PFHxS and female rats were given a dose of 0.5–10 mg/kg PFHxS via IV injection. Each value represents the mean ± SD ($n = 5$–20). *$p < 0.05$ between male and female rats

females, using a UF_H of 10 and UF_S of 10 for human POD. External doses of 0.007 μg/kg/day in male and 0.006 μg/kg/day in female humans were calculated based on the Korean biomonitoring values (male, 2.29 ng/mL; female, 1.75 ng/ml) through the dose metric approach, which is derived from an average plasma concentration from a repeated exposure of PFHxS using the human PBPK model. The MOEs in human males and females were estimated to be 100.5 and 27.7, respectively. Thus, the MOE of human males was about 3.6 times higher than that of females.

Discussion

Sundstrom et al. (2012) reported on the PK parameters of PFHxS in rats after IV or oral administration at a dose of 10 mg/kg with a follow up of 24 h. In female rats, the $t_{1/2}$ (43.9 h) was similar to the result of this study; the CL (4.96 mL/h/kg) was about 2.2-fold higher than ours. In a previous study, we suggested that the gender difference for PFHxS in rats was represented on the basis of PK parameters with $AUC_{0-\infty}$, T_{max}, $t_{1/2}$, and CL (Kim et al. 2016). This study reported that PFHxS was more rapidly absorbed in female rats with a T_{max} of 1.37 h than in male rats (T_{max} = 74.6 h) and was more rapidly eliminated in female rats ($t_{1/2}$ = 41.3 h) than in male rats ($t_{1/2}$ = 645.6 h). In humans, the $t_{1/2}$ between male (35 years) and female (7.7 years) showed the gender differences from paired blood and urine samples (Zhang et al. 2013). On the contrary, Olsen et al. (2007) reported no significant gender differences in humans. However, their study was based on 24 male and 2 female retirees who had worked for at least 20 years in fluorochemical production. We believe that a female sample of only two persons might not provide a reliable indication of gender differences in humans.

In a previous report, the tissue distribution for PFHxS in rats was determined only for the liver (Sundstrom et al. 2012). Therefore, the results for other tissues in this current study could not be compared to those of the previous publication. We investigated the tissue distribution of PFHxS in male and female rats in tissues of the brain, heart, lung,

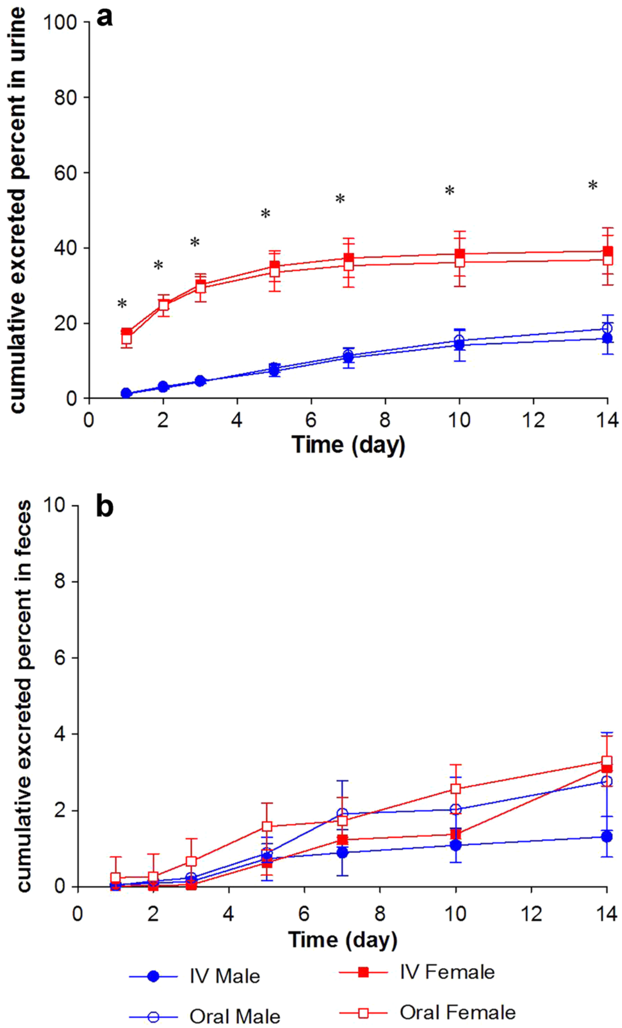

Fig. 6 Cumulative excreted percent (%) of PFHxS in **a** urine and **b** feces. The male rats were administered a dose of 10 mg/kg PFHxS and the female rats were given a dose of 4 mg/kg PFHxS via IV or oral. Each value represents the mean ± SD ($n=5$). *$p<0.05$ between male and female

kidney, liver, spleen, GI tract, adipose tissue, and muscle in order to develop the PBPK model in rats. Among them, PFHxS showed the highest value in the liver. This could be because PFOA and PFOS interfere with the fatty acids binding by the liver fatty acid binding protein (L-FABP) (Luebker et al. 2002). Although PFHxS was known to be associated with ADHD and neurotoxicity, it was detected in the brain at the lowest level. Because PFHxS was highly bound to the plasma protein, only a small amount of free-PFHxS was transferred to the organs (Jing et al. 2009; Kerstner-Wood et al. 2003). Here, the K_p values for PFHxS were < 1, and the PK parameters at doses of 0.5–10 mg/kg for female rats were linear; it could be assumed that the PK parameters for male rats would also be linear. Moreover, the K_p values showed no significant differences according to the administered dose. Therefore, we showed the mean K_p value for all given doses (Fig. 5). The K_p values for PFHxS in male rats were approximately twofold higher than those of the female rats, and the gender differences between the male and female rats were significant. There was approximately a twofold difference of V_d between the male (314.83 ± 23.21 mL/kg) and female rats (163.70 ± 20.28 mL/kg).

Han et al. (2012) classified the renal organic anion transport of human, rat, and mouse kidneys according to the basolateral or apical membrane of proximal tubular cells. The Oat1 and Oat3 were situated on the basolateral membrane of the rat kidney, and Oat2 and Oatp1a1 were known to be on the apical membrane. The Mrp2 was in the direction of the efflux on the apical membrane (Han et al. 2012). Oat1 and Oat3 play important roles in secreting PFASs, while Oatp1a1 plays a major role in the reabsorption of PFASs and is more distributed in male rats (Weaver et al. 2010). In female rats, PFOA was known to be rapidly excreted by Oat1 and Oat3, whereas it was restrictively reabsorbed by Oatp1a1 expression of a low level (Loccisano et al. 2013). Therefore, the PFOA showed a short $t_{1/2}$ and rapid elimination in the female rat. Oatp1a1, which plays a role in reabsorption, is more abundant in male rats than in female rats; PFOA is, therefore, slowly eliminated and has a long $t_{1/2}$ in

Table 3 Biochemical parameters for PBPK modeling of PFHxS in rats and humans

Parameters	Description	Rat		Human	
		Male	Female	Male	Female
T_m	Transport maximum (μg/h)	85,300	910	85,300	910
K_t	Transport affinity constant (μg/L)	15,851	15,851	15,851	15,851
K_u	Rate constant to urine (h^{-1})	0.250	0.591	0.0611[a]	0.144[a]
K_f	Transfer rate constant from GI tract to fecal elimination (h^{-1})	0.02	36.8	0.0049[a]	8.99[a]
K_{abs}	Oral absorption rate constant (h^{-1})	3.73	3.73	0.912[a]	0.912[a]
F_r	Free fraction of PFHxS in blood	0.00076	0.00069	0.00025	0.00023
K_{st}	Rate constant to storage compartment (h^{-1})	10.5	115.7	2.57[a]	28.28[a]

[a] K_{human} means the rate constant in humans and is calculated from Eq. (4)

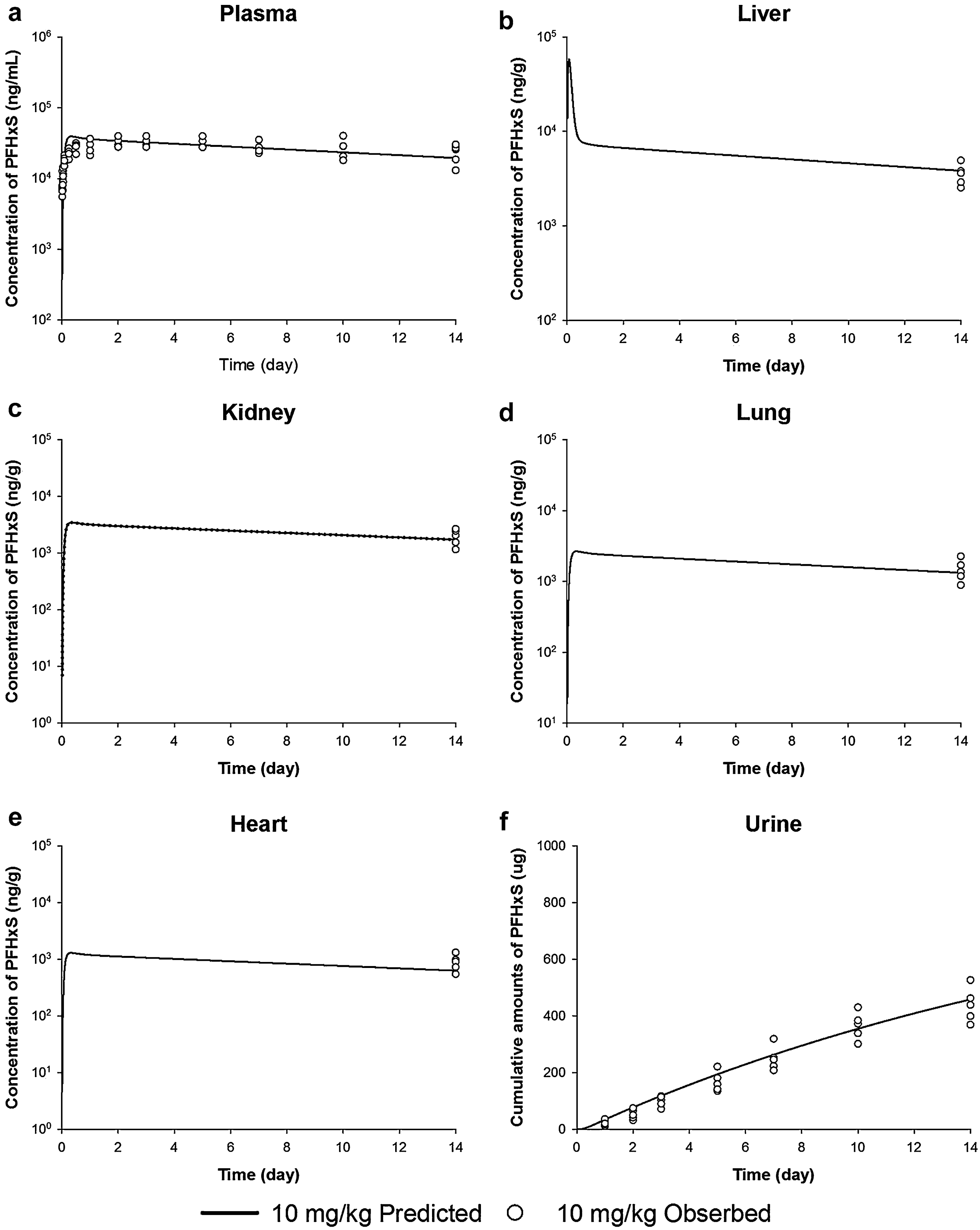

Fig. 7 Predicted (line) and observed (circle) concentration of PFHxS in plasma (**a**), liver (**b**), kidney (**c**), lung (**d**), heart (**e**), and urine (**f**) after oral administration of PFHxS at doses of 10 mg/kg in male rats

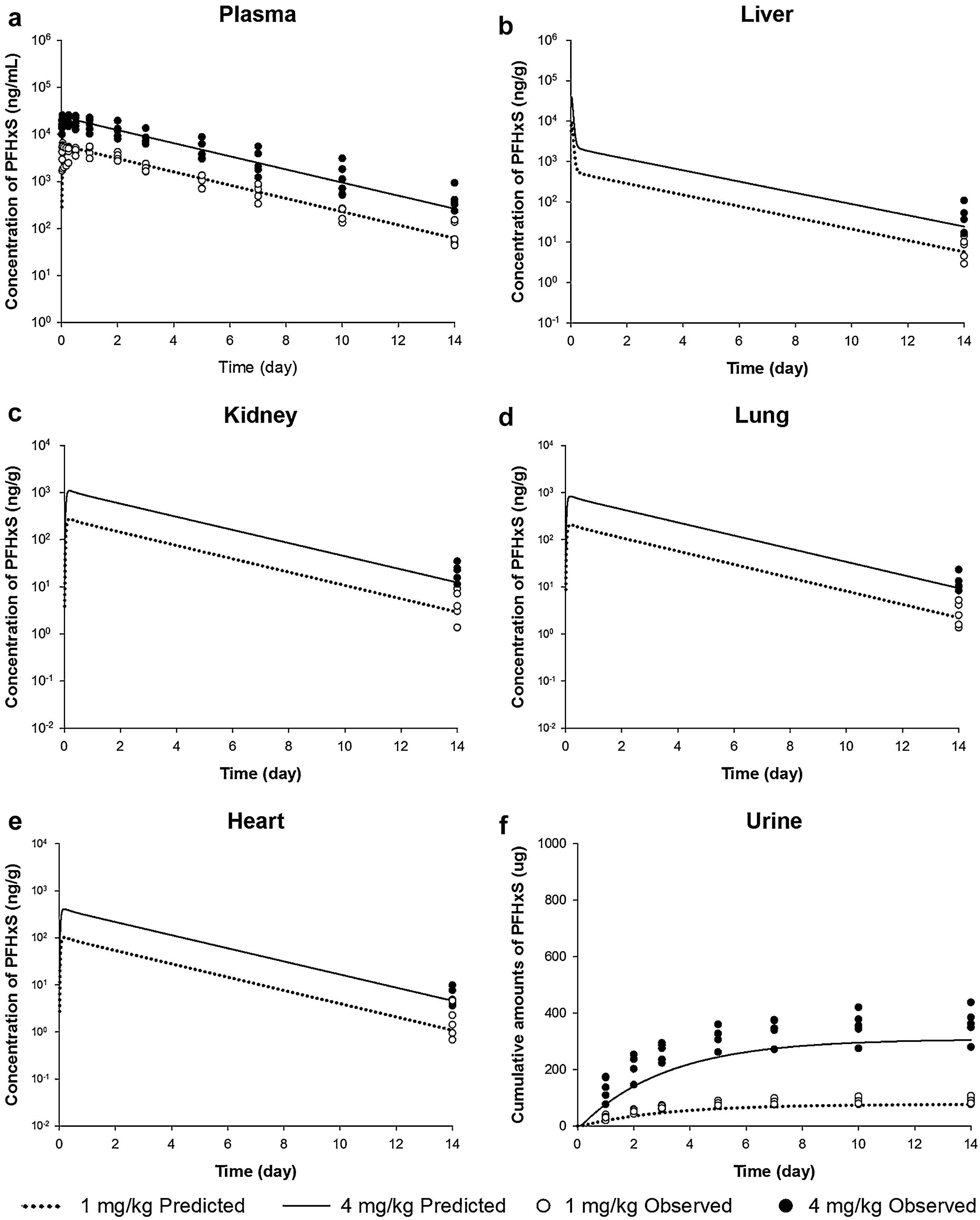

Fig. 8 Predicted (line) and observed (circle) concentration of PFHxS in plasma (**a**), liver (**b**), kidney (**c**), lung (**d**), heart (**e**), and urine (**f**) after oral administration of PFHxS at doses of 1 and 4 mg/kg in female rats

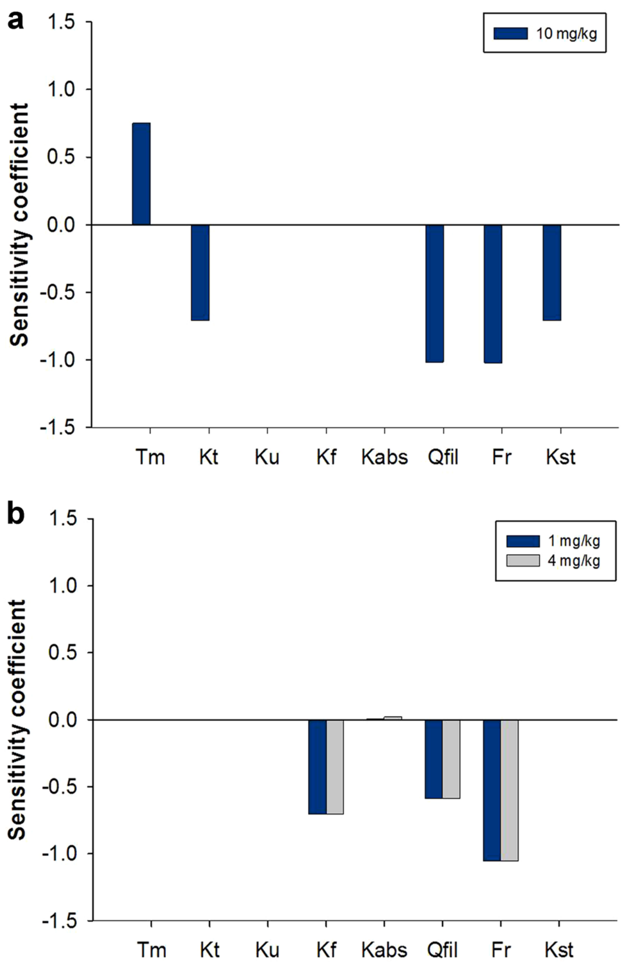

Fig. 9 Sensitivity coefficients for PBPK model parameters for PFHxS in male (**a**) and female (**b**) rats with respect to serum PFHxS AUC

male rats. PFHxS, which has a structure similar to that of PFOA, is predicted to show a gender difference for the same reason as that mentioned above. However, the mechanism of PFHxS for Oat1, Oat3, and Oatp1a1 has not yet been clarified. Moreover, the reabsorption of PFHxS in female rats might be lower than that in male rats due to the low expression of Oat1a1 in female rats. Therefore, the cumulative excreted percentage of PFHxS in the urine of female rats was higher than that of male rats. The linearity of the PK parameters was also applied to explain the cumulative excretion rates of urine and feces. PFHxS excretion through feces after IV administration indicates biliary excretion. In addition, the OATP-mediated uptake of PFOA in the apical membranes of Caco-2 cells could explain the rapid absorption of PFOA (Kimura et al. 2017).

In the sensitivity analysis, while the PBPK parameters of T_m (0.75), K_t (− 0.71), Q_{fil} (− 1.02), F_r (− 1.02) and K_{st} (− 0.71) had a strong influence on the disposition of PFHxS in male rats, they depended on K_f (− 0.70), Q_{fil} (− 0.59) and F_r (− 1.05) in female rats (K_u and K_{abs} with less than 0.01 were not sensitive parameters for the PBPK model without distinction of gender). PFHxS was commonly dependent on the Q_{fil} and F_r in male and female rats. Q_{fil} will still govern the rate at which PFHxS travels from the blood to the filtrate compartment and is then eliminated. The results correspond to PFOA, which is known to be the gender difference (Loccisano et al. 2012a). Loccisano et al. (2012a) reported that PFOA was dependent on the T_m and K_t in male rats, whereas female rats were not sensitive to these parameters which were responsible for renal reabsorption. This could be the reason for the longer $t_{1/2}$ of PFOA in the male rats than in the female rats, and demonstrates that the Oatp1a1, which has a role of reabsorption, was highly distributed in male rats. Unlike PFOA, PFOS was influenced by the parameters of T_m and K_t in male and female rats (Loccisano et al. 2012a). This might explain why the $t_{1/2}$ of PFOS showed no gender difference in the male and female rats (Kim et al. 2016). In addition, although PFOA and PFOS were influenced by the plasma free fraction of chemicals, this parameter was estimated by fitting the plasma concentration data in the PBPK model (Loccisano et al. 2012a). In this study, we used the experimental data for the plasma free fraction of PFHxS via the plasma protein binding assay to develop a more precise PBPK model.

Most PBPK models for PFASs have been used to report on PFOA and PFOS. The development of the PBPK model for PFOA and PFOS in monkeys and the extrapolation to humans were reported by Loccisano et al. (2011). Subsequently, they developed PBPK models for mature male and female rats to characterize the PKs of PFOA and PFOS (Loccisano et al. 2012a). Their model structure consisted of plasma, liver, kidney, and filtrate compartments, as well as a compartment for the remaining body tissues. We elaborated on this simple model structure by adding the heart, lung, adipose tissue, and brain compartments. By adding these compartments, our PBPK model describes the kinetics of PFHxS in the body in greater detail.

Gleason et al. (2015) and Wong et al. (2014) reported that menstruation was a specific elimination route for PFOS in female humans. In previous publications, the structure of the PBPK model for PFASs (PFOA and PFOS) included the blood loss by menstruation to explain the epidemiologic association between serum PFASs levels and early onset of menopause (Ruark et al. 2017), as well as delayed menarche (Wu et al. 2015). Therefore, we added the menstrual blood loss in the PBPK model and compared the effect of menstruation in female humans through our simulation. We found no significant difference in the simulation results with or without menstrual blood loss. This may be due to the negligibly small amount of menstrual blood loss of 69.4 mL per month (Verner and Longnecker 2015), compared to the total blood volume (1.61%). Verner and Longnecker (2015) estimated the final serum loss by considering the non-blood portion

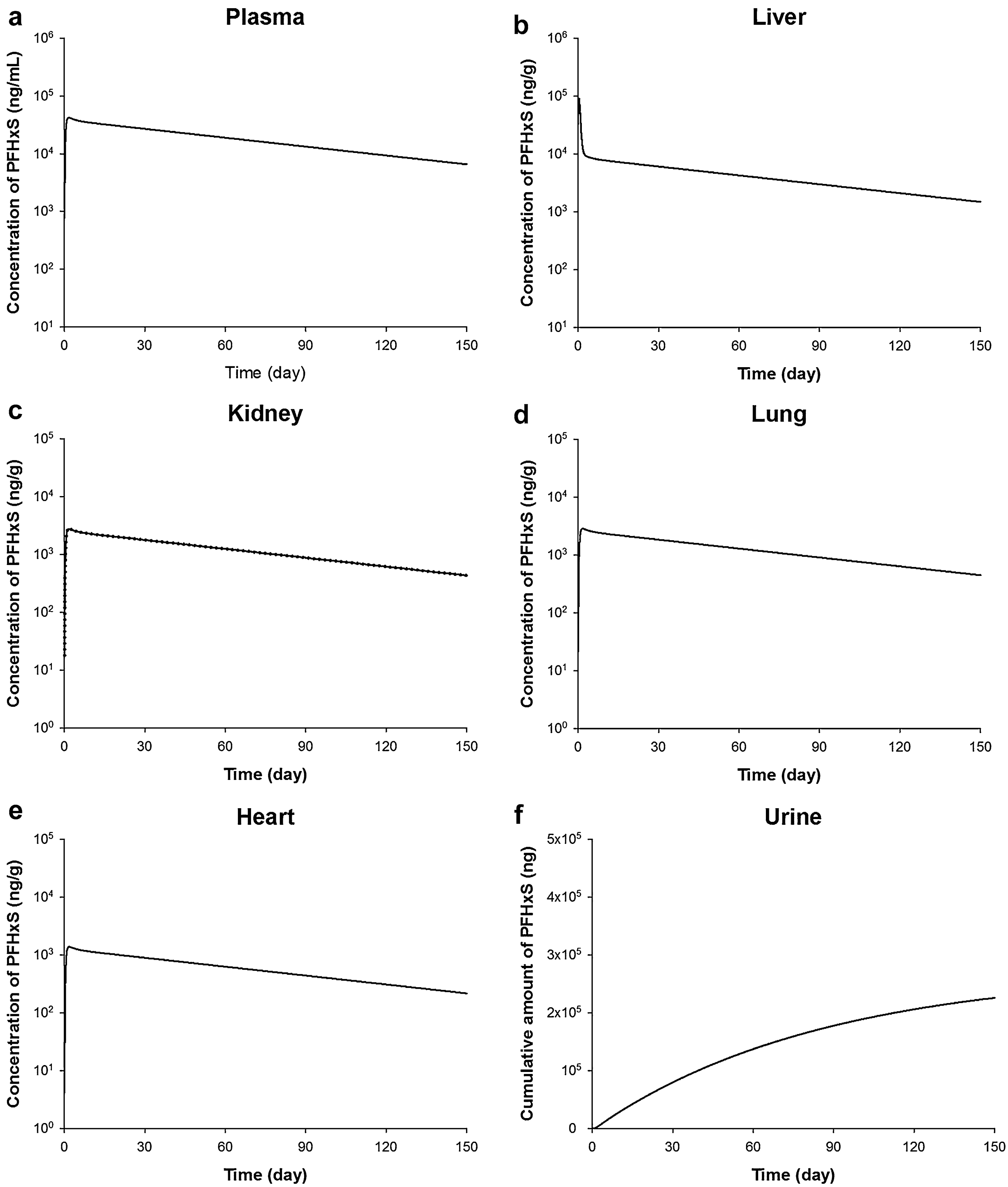

Fig. 10 Predicted concentration of PFHxS in plasma (**a**), liver (**b**), kidney (**c**), lung (**d**), heart (**e**), and urine (**f**) after oral administration of PFHxS at doses of 10 mg/kg in male humans

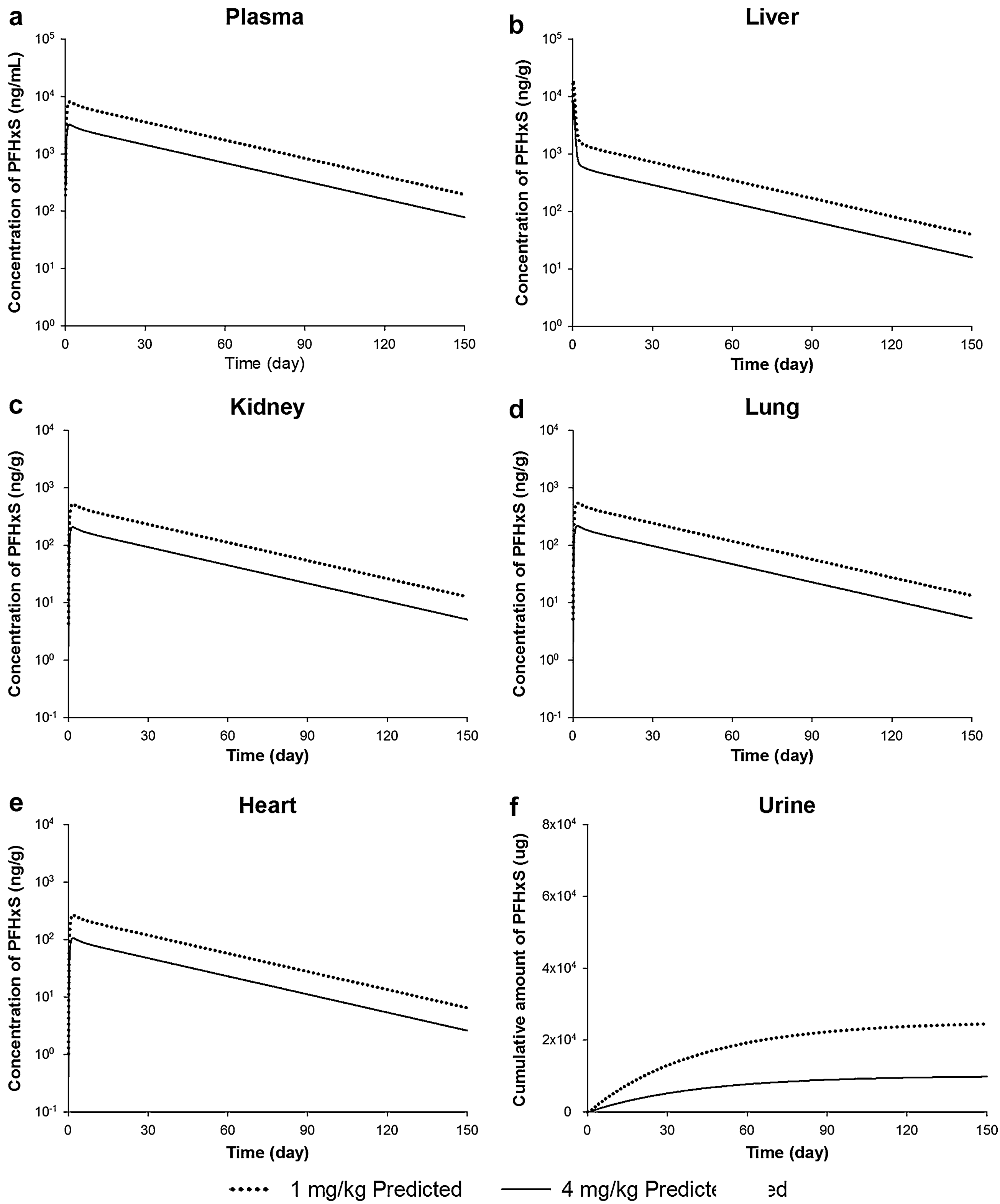

Fig. 11 Predicted concentration of PFHxS in plasma (**a**), liver (**b**), kidney (**c**), lung (**d**), heart (**e**), and urine (**f**) after oral administration of PFHxS at doses of 1 and 4 mg/kg in female humans

of menstrual fluid, albumin concentration (half of the volume), and average hematocrit (40%) in a reproductive-aged woman, based on the population data on menstrual blood volume per cycle (43.4 mL). Based on these data, we can conclude that the menstruation has no direct effect on the sex differences. In the case of female rats, menstruation has not been evaluated. Thus, the menstrual blood loss was only applicable to female humans.

In a comparison of the chemical parameters of PFASs in the PBPK modeling, the T_m of PFOA, which was known to have a gender difference, showed different values with 95,459 µg/h for male rats and 1060.7 µg/h for female rats, and the T_m of PFOS (42,426 µg/h) was the same in male and female rats (Loccisano et al. 2012a). In our study, the T_m of PFHxS was estimated to be 85,300 and 910 µg/h in male and female rats, respectively, and showed a gender difference similar to that of PFOA. The T_m for PFHxS and PFOA in male rats was higher than that in female rats. Loccisano et al. (2012a) reported that the K_t of PFOA and PFOS was 696.8 µg/h and 16,700 µg/h, respectively. When compared with our results (K_t = 15,851 µg/h), the high value of K_t was indicated at a longer $t_{1/2}$. The $t_{1/2}$ was reported to be 1.64 and 0.19 days for PFOA and 20.7 and 0.88 days for PFHxS in male and female rats, respectively, and 26.4 days for PFOS in both genders (Kim et al. 2016). The free fraction of chemicals in the plasma is assumed to be reabsorbed in the filtrate compartment to the kidney (Andersen et al. 2006; Tan et al. 2008) or eliminated from the filtrate compartment to the storage compartment and subsequently to the urine (Loccisano et al. 2011). Loccisano et al. (2012a) reported that the free fraction of PFOA in female rats (0.045) is approximately seven times higher than that of male rats (0.006) based on the simulation results. In the case of PFOS, which is known to have no gender difference, the same value of free fraction (0.022) was shown in both genders. On the contrary, in the present study, the free fraction for PFHxS was estimated by the plasma protein binding assay and showed no significant gender differences (0.00076 in male and 0.00069 in female rats; 0.00025 in male and 0.00023 in female humans). The results showed a difference of about three times between rats and humans. The free fraction seems to be an unsuitable parameter to describe the gender differences for PFASs.

Liu et al. (2015) conducted a risk assessment for 11 PFASs including PFHxS in the atmosphere of Shenzhen, China. This report was used to establish the hazard ratio for non-cancer risk through breathing based on PFOS (HQ, 3.5E-6) and PFOA (2.4E-7) concentrations. The PFHxS was excluded in the risk assessment because of its relatively low concentration in Shenzhen. In the general population of Sweden, no concern was shown for hepatotoxicity (HQ, 0.0048) and reproductive toxicity (HQ, 0.0035) due to PFHxS. The HQs were estimated from the results of actual biomonitoring and animal data of serum levels obtained from the literature. An HQ of less than 1 means an absence of risk for the considered endpoint, while an HQ greater than 1 indicates unsafe exposure (safe $1 < HQ < 1$ unsafe) (Borg et al. 2013). Meanwhile, we calculated the MOE and RfD according to the guidance on the risk assessment using the PBPK model established by WHO (2010). In 2016, US EPA released the RfDs which are ranged from 20 to 150 ng/kg/day for PFOA and 20–50 ng/kg/day for PFOS regardless of gender (Dong et al. 2017). In our study, the RfD (159 ng/kg/day) of PFHxS in female was similar with that of PFOA based on the rat POD (150 ng/kg/day) for liver weight gain and necrosis. Borg et al. (2013) suggested that the RfDs of PFHxS were 98–196 ng/mL for the hematology. Also, the higher value of MOE indicates no concern in the health risk assessment. The MOEs in male (100.5) and female (27.7) humans might not be of concern. There have been several reports on PFOA and PFOS whereas reports of PFHxS are relatively rare, and a PBPK model of PFHxS that considers the sex difference has not been reported to date. Therefore, we developed and evaluated a PBPK model for PFHxS while considering the gender difference between male and female rats. We then used the PBPK model for a human risk assessment under a recommendation made by WHO (2010) of the way in which the PBPK model could be used in the risk assessment (WHO 2010). The PBPK model could be used as a rational and ethical approach for assessing human health risks resulting from environmental exposure of toxic chemicals, as an alternative to using the limited data of PKs.

Conclusion

In conclusion, the PBPK model was adequately developed and evaluated for PFHxS based on a PK study and an analysis of tissue distribution after oral or IV administration in rats. The developed PBPK model used in male and female rats provides a foundation to extrapolate the PBPK model from rats to male and female humans. The PBPK model for PFHxS was shown to be helpful for human health risk assessment in Korean populations regarding the potential for accumulated health problems in humans exposed to PFHxS. We found that the detected serum levels of PFHxS in Korean men and women are not a concern at present.

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Compliance with ethical standards

Conflict of interest The authors have declared no conflict of interest.

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